Physics 9702 Paper 4 · Topic 14

Temperature

The seven hardest A Level Structured Questions on this topic from the 2020 to 2026 papers, hardest first, followed by their mark schemes.

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1 9702/41 May/June 2025, Question 3 (9 marks) page 2

Examiner report: (b) A comparison of the change in volume or molecular separation and the corresponding change in potential energy of the molecules between vaporisation and fusion was rarely explained well by candidates. Very few candidates who had an explanation in terms of volume and potential energy change referred to the kinetic energy of molecules being unchanged.

2 9702/42 February/March 2025, Question 3 (12 marks) page 3

Examiner report: However, only the strongest candidates correctly analysed the energy transfers involved to arrive at the correct answer. However, only the strongest candidates realised that, because the gas is expanding, the work done on the gas must be negative.

3 9702/42 October/November 2024, Question 3 (11 marks) page 6

Examiner report: Weaker candidates found it difficult to work out the mass of the substance, with use of the volume of the gas with the density of the liquid being the common incorrect starting point. (c) Only the strongest candidates were able to give a response to this question that went beyond IGCSE level physics.

4 9702/41 May/June 2023, Question 3 (11 marks) page 8

Examiner report: (a) Candidates found it difficult to demonstrate a good understanding of thermal equilibrium. (ii) This calculation proved to be challenging.

5 9702/41 October/November 2022, Question 2 (10 marks) page 10

Examiner report: (c) (i) This question proved to be challenging.

6 9702/42 October/November 2025, Question 3 (9 marks) page 12

Examiner report: (iii) Candidates found this question difficult. Many candidates did not appreciate the significance of the information in the question, that the system needs to reach equilibrium before the measurement can be made.

7 9702/42 February/March 2026, Question 3 (8 marks) page 14

Examiner report: Candidates often omitted one of the terms in the conservation of energy application, usually forgetting that the ice undergoes two stages of warming (one from –18 °C to 0 °C and the other from 0 °C to the final temperature).

Questions were chosen using the Cambridge examiner reports (Principal Examiner Report for Teachers): parts the reports describe as challenging, poorly answered, or answered fully only by the strongest candidates. May/June 2020 and May/June 2026 have no report, so questions from those series were judged directly. Variant 43 repeats variant 41 in every series, so only 41 is cited.

3 **(a)** Define specific latent heat.

.....

.....

..... [2]

(b) Explain why, for a substance, the specific latent heat of vaporisation is usually greater than the specific latent heat of fusion.

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..... [3]

(c) An ice cube of mass 37.0 g at temperature 0.0 °C is placed in a beaker containing water of mass 208 g at temperature 26.4 °C.

When all the ice has melted, and all the water in the beaker has reached thermal equilibrium, the final temperature of all the water is 10.3 °C.

The specific heat capacity of water is 4.18 $J\,g^{-1}\,°C^{-1}$.

The beaker has negligible specific heat capacity and is perfectly insulated from the surroundings.

Determine a value, to three significant figures, for the specific latent heat of fusion of water.

specific latent heat of fusion = ... $J\,g^{-1}$ [4]

[Total: 9]

3 **(a)** Two metal cuboids P and Q are in thermal contact with each other.

(i) P and Q are in thermal equilibrium.

State what is meant by the term thermal equilibrium.

.....

.....

..... [2]

(ii) Data for P and Q are given in Table 3.1.

Table 3.1

	P	Q
specific heat capacity / J kg^{-1} K^{-1}	390	910
mass / kg	0.54	0.37

P and Q are initially both at the same temperature.

P is supplied with 24 kJ of thermal energy. After some time, P and Q are once again both at the same temperature as each other.

P and Q are perfectly insulated from the surroundings.

Determine the change in temperature ΔT of Q.

ΔT = ... K [3]

(b) Nitrogen may be assumed to be an ideal gas. A fixed amount of nitrogen gas is contained at a constant pressure of 1.6×10^5 Pa.

The variation of the volume V of the gas with the temperature θ of the gas is shown in Fig. 3.1.

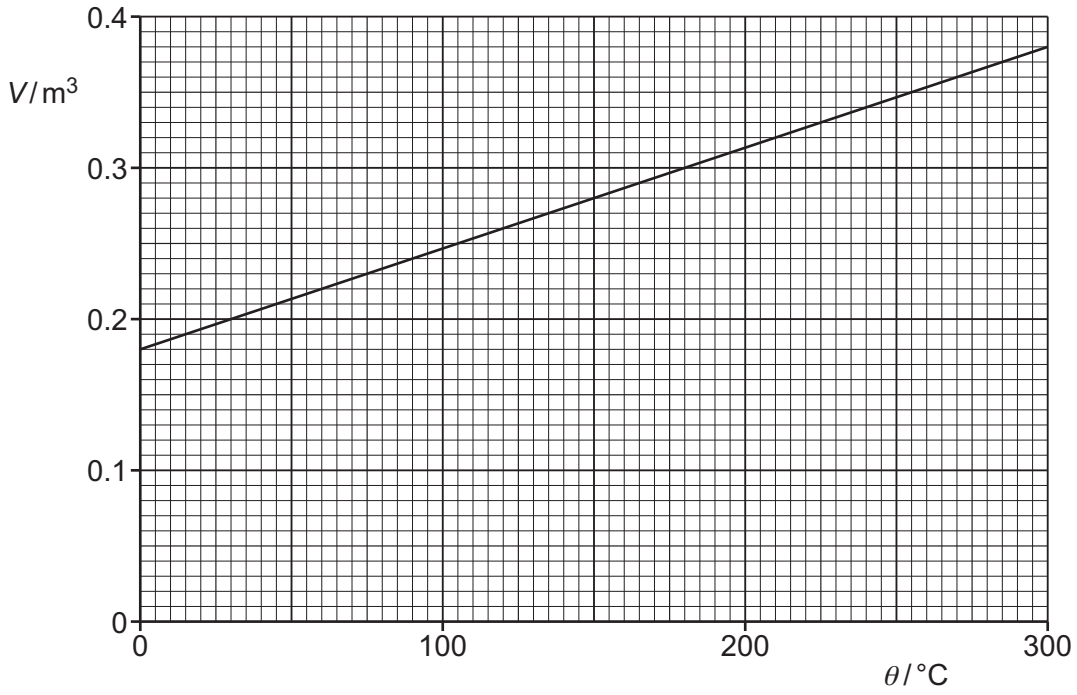


Fig. 3.1

(i) The temperature of the nitrogen gas is increased from 0 °C to 210 °C.
Determine the work done on the gas.

work done = ... J [3]

(ii) Determine the number N of molecules of nitrogen gas.

N = ... [2]

(iii) The mass of a nitrogen molecule is 4.7×10^{-26} kg.

Calculate the root-mean-square (r.m.s.) speed of a nitrogen molecule at 210 °C.

r.m.s. speed = ... $m\,s^{-1}$ [2]

[Total: 12]

3 **(a)** Define specific latent heat.

.....

.....

..... [2]

(b) A dish containing $7.2 \times 10^{-5}\,m^3$ of a substance rests on a laboratory bench. The substance is initially a liquid of density $710\,kg\,m^{-3}$. Atmospheric pressure is $1.0 \times 10^5\,Pa$.

The liquid is heated at its boiling point so that it completely vaporises. The increase in the internal energy of the substance during this process is 17.6 kJ. The final volume of the vapour is $0.017\,m^3$.

(i) Show that the magnitude of the work done on the substance when it vaporises is 1.7 kJ.

[2]

(ii) Use the information in **(b)(i)** to calculate the thermal energy Q, in kJ, supplied to the substance to cause it to vaporise.

Q = ... kJ [2]

(iii) Use your answer in **(b)(ii)** to determine a value for the specific latent heat of vaporisation L_V, in $kJ\,kg^{-1}$, of the substance.

L_V = ... $kJ\,kg^{-1}$ [2]

(c) The substance in **(b)** has a specific latent heat of fusion L_F.

Suggest and explain whether L_F is likely to be less than, the same as, or greater than the answer in **(b)(iii)**.

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.....

.....

..... [3]

[Total: 11]

3 **(a)** State the reason why two objects that are at the same temperature are described as being in thermal equilibrium.

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..... [1]

(b) Fig. 3.1 shows the variations with temperature of the densities of mercury and of water between 0 °C and 100 °C.

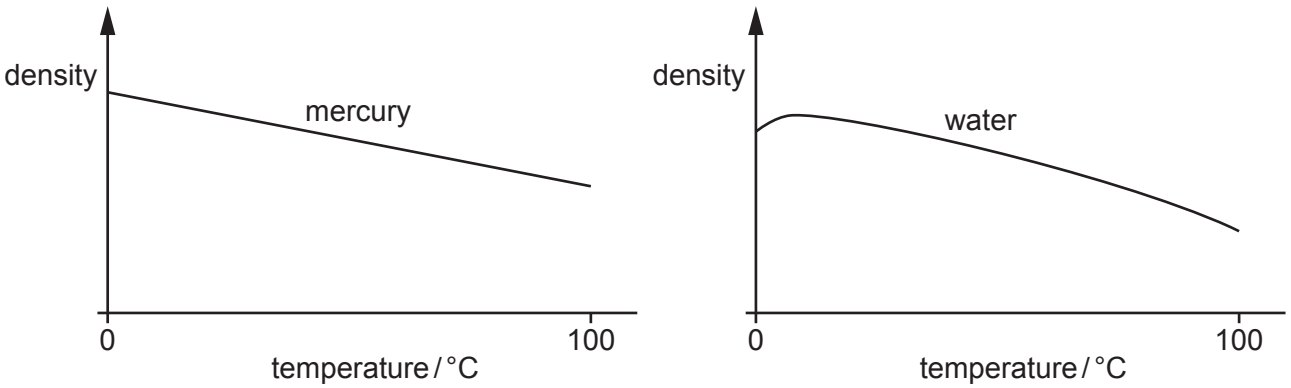


Fig. 3.1

Temperature may be measured using the variation with temperature of the density of a liquid.

Suggest why, for measuring temperature over this temperature range:

(i) mercury is a suitable liquid

.....

..... [1]

(ii) water is not a suitable liquid.

.....

.....

..... [2]

(c) A beaker contains a liquid of mass 120 g. The liquid is supplied with thermal energy at a rate of 810 W. The beaker has a mass of 42 g and a specific heat capacity of 0.84 $J\,g^{-1}\,K^{-1}$. The beaker and the liquid are in thermal equilibrium with each other at all times and are insulated from the surroundings.

Fig. 3.2 shows the variation with time *t* of the temperature of the liquid.

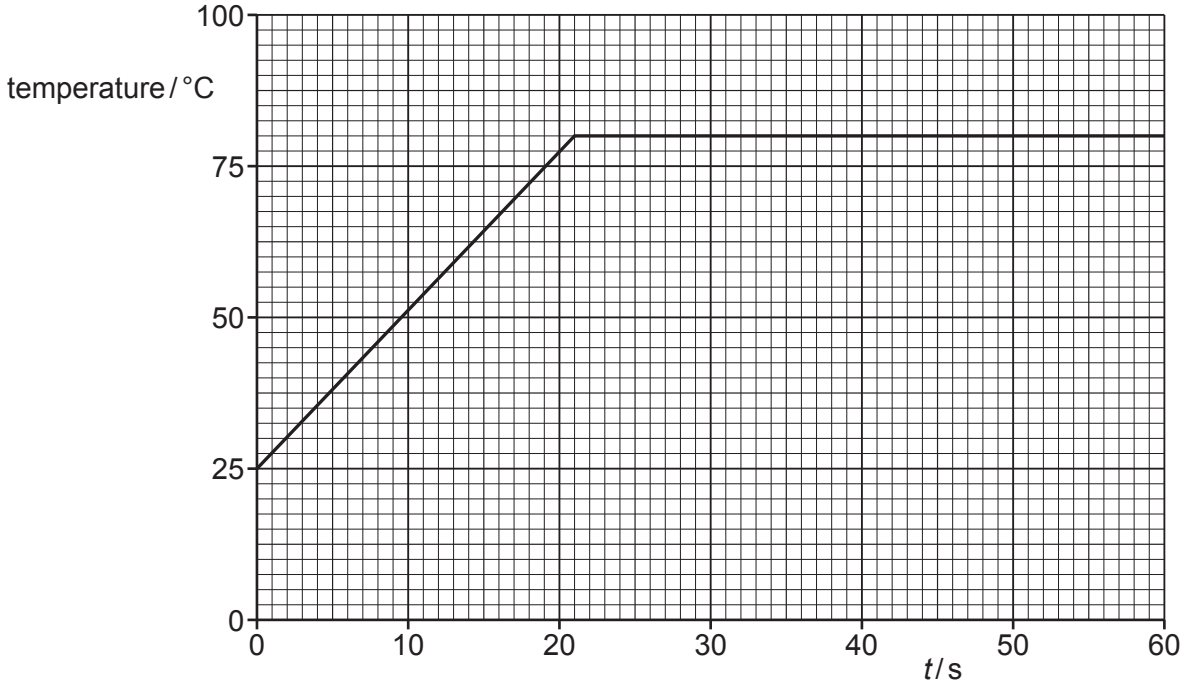


Fig. 3.2

(i) State the boiling temperature, in °C, of the liquid.

temperature = ... °C [1]

(ii) Determine the specific heat capacity, in $J\,g^{-1}\,K^{-1}$, of the liquid.

specific heat capacity = ... $J\,g^{-1}\,K^{-1}$ [4]

(d) The experiment in **(c)** is repeated using water instead of the liquid in **(c)**. The mass of liquid used, the power supplied, and the initial temperature are all unchanged.
The specific heat capacity of water is approximately twice that of the liquid in **(c)**.
The boiling temperature of water is 100 °C.

On Fig. 3.2, sketch the variation with time t of the temperature of the water between $t = 0$ and $t = 60$ s. Numerical calculations are not required. [2]

[Total: 11]

2 Fig. 2.1 shows a laboratory thermometer that is calibrated to measure temperature in degrees Celsius.

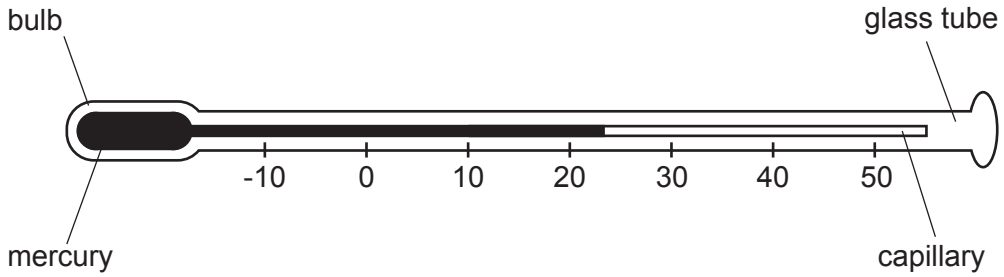


Fig. 2.1

The thermometer makes use of the fact that the density of mercury varies with temperature.

(a) State **two** other physical properties of materials, apart from the density of a liquid, that can be used for measuring temperature.

1 .....

2 .....
[2]

(b) The thermometer is initially at 23.0 °C, as shown in Fig. 2.1. It is used to measure the temperature of an insulated beaker of water that is at 37.4 °C. The bulb of the thermometer is inserted into the water, and the water is stirred until the reading on the thermometer becomes steady.

The mass of water in the beaker is 18.7 g.
The mass of mercury in the thermometer is 6.94 g.
The specific heat capacity of water is 4.18 J g^{-1} K^{-1}.
The specific heat capacity of mercury is 0.140 J g^{-1} K^{-1}.
The glass of the thermometer and the beaker containing the water can be considered to have negligible heat capacity.

(i) Calculate, to three significant figures, the final steady temperature indicated by the thermometer in the water.

temperature = ... °C [4]

(ii) Suggest **one** change that could be made to the design of the thermometer that would enable it to give a more accurate measurement of temperature.

.....

..... [1]

(c) (i) Explain why the thermometer in Fig. 2.1 does **not** provide a direct measurement of thermodynamic temperature.

.....

.....

..... [2]

(ii) Thermodynamic temperature T may be determined by the behaviour of a type of substance for which T is proportional to the product of pressure and volume.

State the name of this type of substance.

..... [1]

[Total: 10]

3 **(a)** State what is meant by two objects being in thermal equilibrium.

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..... [2]

(b) Fig. 3.1 shows a type of thermometer called a constant volume gas thermometer.

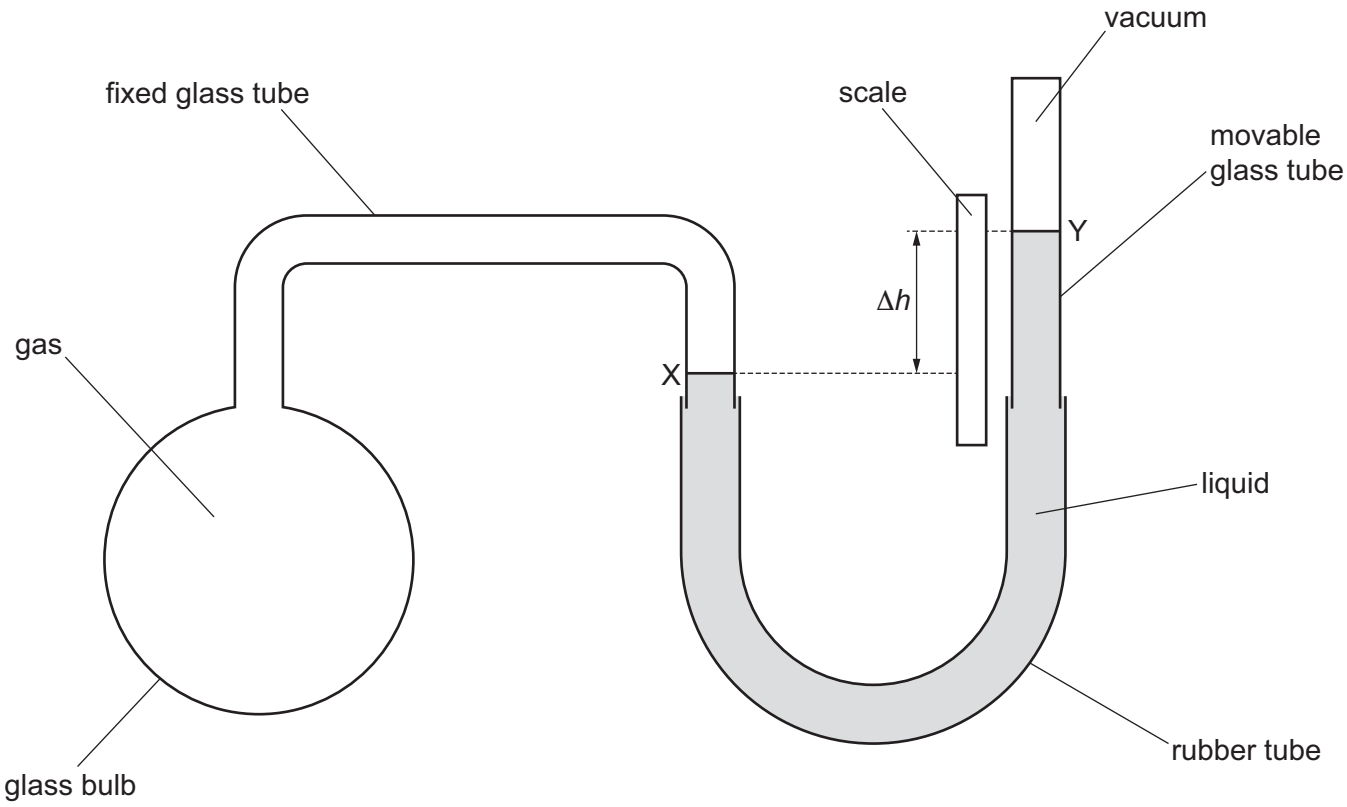


Fig. 3.1 (not to scale)

The thermometer is used to determine the thermodynamic temperature T of the gas in the glass bulb.

The glass bulb is immersed in the environment for which the temperature is to be measured. The height of the movable glass tube is then adjusted so that the level of the liquid on the left-hand side aligns with the reference line X marked on the fixed glass tube. The reference line Y is marked on the side of the movable glass tube. The level of the liquid at Y is higher than at X as a result of the pressure of the gas in the glass bulb.

The difference in height Δh between the liquid levels at X and Y is then measured using the scale. The thermodynamic temperature T of the gas is directly proportional to the pressure of the gas. This pressure is directly proportional to Δh.

(i) The value of Δh can be used to calculate the pressure of the gas. In order to do this, the gravitational field strength is used, along with a property of the liquid.

State the property of the liquid that is used to calculate the pressure.

..... [1]

(ii) Before the measurement of Δh can be made, the glass bulb needs to reach thermal equilibrium with the environment for which the temperature is to be measured.

State **two** disadvantages of using a constant volume gas thermometer to measure temperature.

1 .....

.....

2 .....

.....
[2]

(iii) Suggest **one** situation in which a constant volume gas thermometer would be an appropriate type of thermometer to choose for measuring temperature.

.....

.....

..... [1]

(iv) Level X aligns with 2.31 cm on the scale. At 0 °C, level Y aligns with 8.69 cm.

At temperature θ, level Y aligns with 7.83 cm on the scale.

Determine a value for θ in °C.

θ = ... °C [3]

[Total: 9]

3 **(a)** Define specific latent heat of fusion.

.....

.....

..... [2]

(b) Explain why substances usually have a higher specific latent heat for vaporisation than for fusion.

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.....

.....

..... [2]

(c) An ice cube of mass 15.6 g at a temperature of –18 °C is removed from a freezer.

The ice cube is immediately placed in an insulated beaker that contains a mass of 132.4 g of water at a temperature of 34 °C.

The specific heat capacity of water is 4.19 kJ kg^{-1} °C $^{-1}$.
The specific heat capacity of ice is 2.11 kJ kg^{-1} °C $^{-1}$.
The specific latent heat of fusion of water is 336 kJ kg^{-1}.
The beaker has negligible specific heat capacity.
The contents of the beaker are left to reach thermal equilibrium.

Calculate the temperature, in °C, of the water in the beaker when thermal equilibrium is reached.

temperature = ... °C [4]

[Total: 8]

Question	Answer	Marks
3(a)	(thermal) energy per unit mass (to cause state change)	**B1**
	(thermal) energy to change state at constant temperature	**B1**
3(b)	(for vaporisation): involves greater change in volume (of substance) **or** involves greater increase in separation of molecules	**B1**
	more work has to be done by molecules (to separate) **or** greater increase in potential energy of molecules	**M1**
	kinetic energy of molecules unchanged, so more thermal energy needed	**A1**
3(c)	$Q = mc\Delta\theta$ **and** $Q = mL$	**C1**
	$\Delta\theta$ for the water = 26.4 – 10.3	**C1**
	$(37.0 \times L) + (37.0 \times 4.18 \times 10.3) = (208 \times 4.18 \times 16.1)$	**C1**
	$L = 335\ \text{J g}^{-1}$	**A1**

Question	Answer	Marks
3(a)(i)	(P and Q are at the) same temperature	**B1**
	no net transfer of thermal energy (between P and Q)	**B1**
3(a)(ii)	$Q = mc\Delta T$	**C1**
	$24 \times 10^3 = (0.54 \times 390 \times \Delta T) + (0.37 \times 910 \times \Delta T)$	**C1**
	$\Delta T = 44\,\text{K}$	**A1**
3(b)(i)	work done $= p\Delta V$	**C1**
	$= (1.6 \times 10^5) \times (0.18 - 0.32)$	**C1**
	$= -2.2 \times 10^4\,\text{J}$	**A1**
3(b)(ii)	$pV = NkT$	**C1**
	$N = (1.6 \times 10^5 \times 0.18) / (1.38 \times 10^{-23} \times 273)$ $= 7.6 \times 10^{24}$	**A1**
3(b)(iii)	$\frac{1}{2}m\langle c^2\rangle = (3/2)kT$	**C1**
	r.m.s. speed $= \sqrt{[(3 \times 1.38 \times 10^{-23} \times (210 + 273) / (4.7 \times 10^{-26})]}$ $= 650\,\text{ms}^{-1}$	**A1**

Question	Answer	Marks
3(a)	(thermal) energy per unit mass (to cause change of state)	**B1**
	(thermal) energy to change state at constant temperature	**B1**
3(b)(i)	$W = p\Delta V$	**C1**
	$= 1.0 \times 10^5 \times 0.017 = 1700$ J $= 1.7$ kJ	**A1**
3(b)(ii)	$\Delta U = Q + W$	**C1**
	$Q = 17.6 + 1.7$ $= 19.3$ kJ	**A1**
3(b)(iii)	mass $= 710 \times 7.2 \times 10^{-5}$ ($= 0.051$ kg)	**C1**
	$L = 19.3 / 0.051$ $= 380$ kJ kg^{-1}	**A1**
3(c)	fusion involves (much) smaller volume change (than vaporisation)	**B1**
	smaller change in intermolecular spacing so smaller change in internal energy	**B1**
	negligible work done (by substance during fusion) so L_F is less (than L_V)	**B1**

Question	Answer	Marks
3(a)	no net thermal energy is transferred (between them)	**B1**
3(b)(i)	variation (of density with temperature) is linear **or** each temperature has a unique value of density	**B1**
3(b)(ii)	• variation (of density with temperature) is not linear • region where the density does not vary with temperature • different temperatures have the same density *Any two points, 1 mark each*	**B2**
3(c)(i)	boiling point = 80 °C	**A1**
3(c)(ii)	$Q = Pt$ **and** $t = 21$ s (thermal energy supplied = $810 \times 21 = 17000$ J)	**C1**
	$c = Q / m\Delta\theta$	**C1**
	thermal energy absorbed by beaker = $42 \times 0.84 \times (80 - 25)$ (= 1940 J)	**C1**
	s.h.c. of liquid = $[(810 \times 21) - (42 \times 0.84 \times (80 - 25))] / [120 \times (80 - 25)]$ = 2.3 J g^{-1} K^{-1}	**A1**
3(d)	sketch: straight diagonal line from 25 °C to 100 °C **and** then horizontal at 100 °C	**B1**
	straight diagonal line starting at 25 °C with gradient approximately half that of the original line	**B1**

Question	Answer	Marks
2(a)	• resistance of a metal • volume of a gas at constant pressure • e.m.f. of a thermocouple *Any two points, 1 mark each*	**B2**
2(b)(i)	$Q = mc\Delta T$	**C1**
	evidence of realisation that Q lost by water = Q gained by mercury	**C1**
	$18.7 \times 4.18 \times (37.4 - T) = 6.94 \times 0.140 \times (T - 23.0)$	**C1**
	$T = 37.2$ °C	**A1**
2(b)(ii)	use a liquid with a lower (specific) heat capacity (than mercury) **or** use a smaller mass of mercury	**B1**
2(c)(i)	depends on properties of a real substance	**B1**
	0 °C is not absolute zero	**B1**
2(c)(ii)	ideal gas	**B1**

Question	Answer	Marks
3(a)	same temperature	**B1**
	no net transfer of thermal energy (between them)	**B1**
3(b)(i)	density	**B1**
3(b)(ii)	*Any two points from:* • large response time / large time to reach equilibrium **or** cannot measure rapidly changing temperatures • reaching equilibrium requires (significant) transfer of energy **or** changes temperature of environment being measured **or** cannot measure temperature of small objects • bulky / difficult to set up **or** difficult to take readings / scale not calibrated to read temperature **or** cannot measure temperature of solid objects	**B2**
3(b)(iii)	substance with large mass **or** temperature that is constant (over time) **or** to calibrate other thermometers (in a laboratory)	**B1**

Question	Answer	Marks
3(b)(iv)	0 °C = 273 K	**C1**
	T = 273 × (7.83 – 2.31) / (8.69 – 2.31) (= 236 K)	**C1**
	θ = 236 – 273 = – 37 °C	**A1**

Question	Answer	Marks
3(a)	thermal energy per unit mass (to cause change of state)	**B1**
	(thermal) energy required to cause change of state between solid and liquid without change in temperature	**B1**
3(b)	change in volume is greater for vaporisation	**B1**
	either: increase in volume so work is done against the atmosphere *or*: work done against atmosphere so more thermal energy supplied	**B1**

Question	Answer	Marks
3(c)	$Q = mc\Delta\theta$	**C1**
	evidence of correct application of conservation of thermal energy [e.g. loss in energy of water as it cools = gain in energy of original ice as it warms + gain in energy of ice as it melts]	**C1**
	$[(34 - \theta) \times 132.4 \times 4.19] = [(\theta \times 15.6 \times 4.19) + (18 \times 15.6 \times 2.11) + (15.6 \times 336)]$	**C1**
	temperature = **21 °C**	**A1**